

28 period $T = \frac{100}{2} = 50 \text{ ms}$

$$f = \frac{1}{50 \times 10^{-3}} = 20 \text{ Hz}$$

$$P_{\text{max}} = I_o^2 R$$

$$I_o = \sqrt{\frac{50}{10}} = 2.24 \text{ A}$$

$$I_{r.m.s.} = \frac{2.24}{\sqrt{2}} = 1.58 \approx 1.6 \text{ A}$$

Answer: A

29 For diffraction of the electron beam, the de Broglie wavelength is $\lambda = \frac{h}{mv} = 10^{-10} \text{ m}$